

Date: 21 Jan 2025

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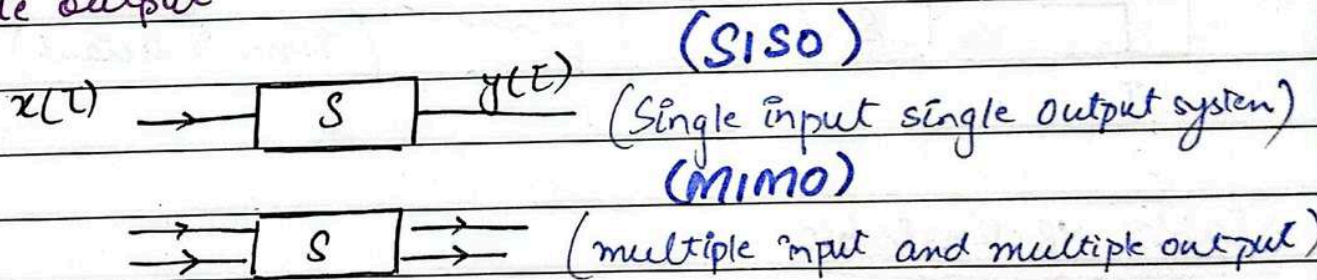
Feedback Control System

Lecture 1

Book: Control system engineering 6th edition
Norman S. Niece

Reference Book: Modern control Engineering 5th edition
Katsuhiko Ogata

- Convolution: we perform convolution to get the output of specific system
- System is a set of interconnections which are connected together we apply some input to that system process that input converts it either in a different form or keeps it in same form and generates a output
- System is some set of components which takes input process it and generate output

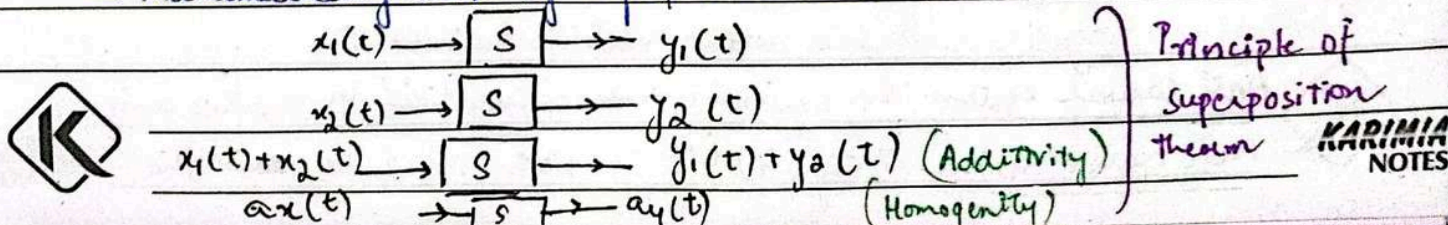


Properties / Types of System:

System can be classified as

→ Linear and Non Linear Systems

Linear system System is linear if it satisfied principle of superposition
↳ Mathematically check by superposition theorem)



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Non Linear if it does not satisfy superposition theorem

All mathematical tools is apply on linear system

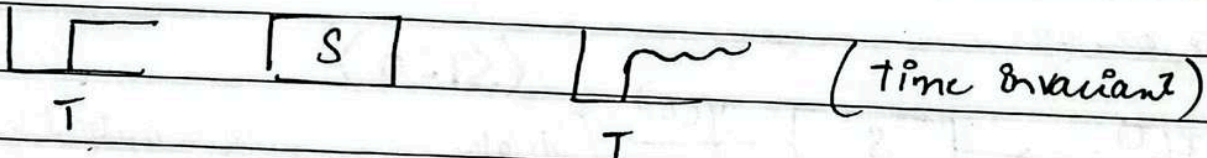
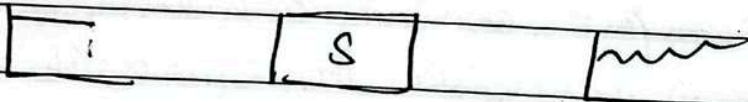
non linearity are unpredictable

Linear device

non linear devices are replace by linear devices or approximation

Time varying and Time invariant

Time invariant if delay in the input cause the same delay in the output
or in other word the characteristics of system does not change with time
example: resistor & capacitor (linear component)
Thermistor is non linear and time varying bcz its value changes depend on surroundings.



Stable vs Unstable

Stable if you provide bounded input you will get bounded output

Causal vs Non causal

Causal System if output depend only on present and past value not on future value

negative axis ma value exist nahi karti



Anti Causal system the system whose value exist in negative axis

Non causal System signal is on both +ve & -ve axis

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Feedback Control System:

Control System: The system which control something according to the requirement

System should define on basis of its functionality

Example Temperature control

Control Base system is a goal base system

In this course system is define on the basis of goal

Application of Control System

→ Main focus is on transfer function approach (frequency domain approach)

→ Another approach will be time domain approach (state space approach)

→ Heating & cooling

→ Flight control

→ Temperature control

→ Robotics

Control system: A system that can regulate itself another system

→ All industrial processes

→ Space craft

A device to manage command to direct or regulate the behaviour of another device or system

Use has a goal and select an input to system to achieve goal

Open Loop system: (No feedback) (no check and balance of output) (normally dependent on time hoping to get the desired output) (Some case you get the output sometimes not) (accuracy is not matter)
Home appliances is example of open loop system.

Closed Loop (output is monitor continuously) (loop of system)



→ AC is example

→ open loop

Both system has goal but fan & AC is different

→ close loop

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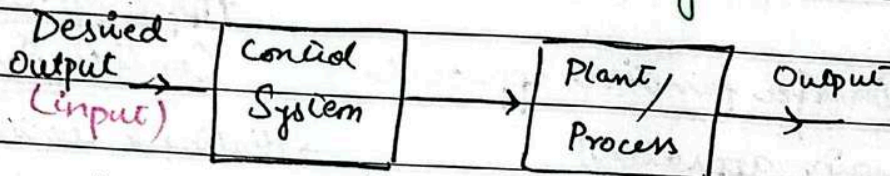
Components (process) (input & output)

↳ take place inside a system to provide input & output

Components of Control System:

General System
↳ Input
↳ Output
↳ want to achieve input & output (objective)
↳ control action (every control system has a control action that is performed by device called controller)

Control System



These both parts are present in open & close loop

→ Motor is called Plant

→ Process based system called process

Open loop

↓
Time is set

↓
in block diagram

↓
No output is monitor

Plant/Process: portion of the system which is to be controlled or regulated

Controller: control plant or process

Input: Desired purpose

Output: Action response (continuously monitor)

Disturbance: (something unwanted that badly affects output)

is a signal that tends to adversely affect the performance

of system. It is unwanted input

→ analysis can be done by keeping linearity in mind

find actual input ki output find disturbance ki output

find then sum up both

Sensor/Transducer: we always do not have electrical signal

we have other signal like temperature we need to convert



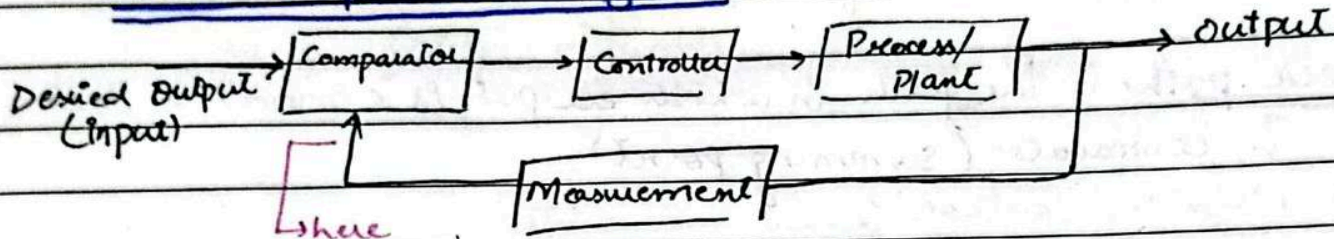
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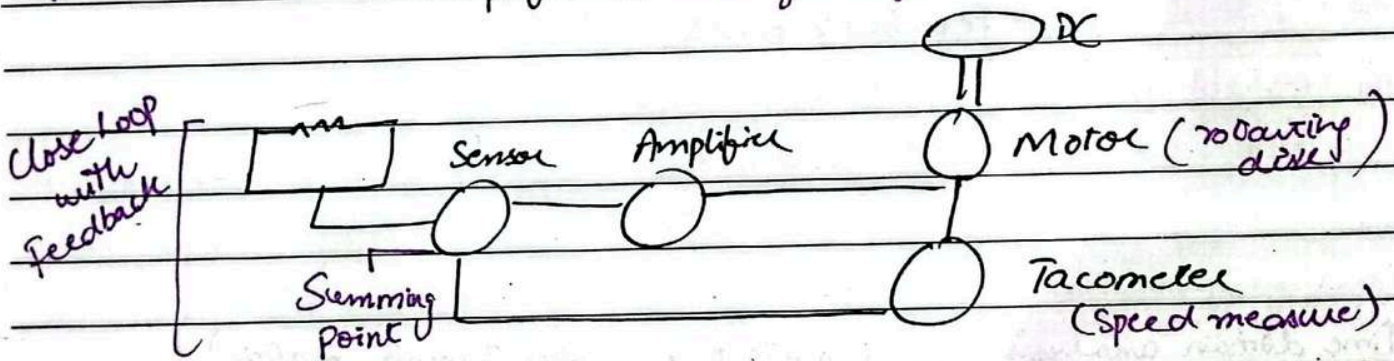
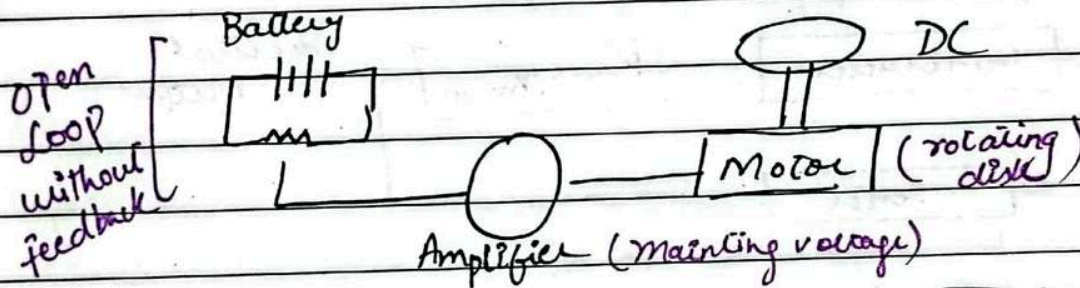
Summing point / Forward path / feedback path

Closed Loop control system:



(Negative feedback we normally don't signal as difference delay)

Comparator $\leftarrow \oplus$



this control system is application to all field of life

Classification of Control System

- ↳ Natural Control System Universe (star, planets)
- ↳ Man Made control system Aerospace
- ↳ Manual control system (doing without machine)
- ↳ Automatic control system (doing with machine)



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Feedback accuracy zada complexity zada

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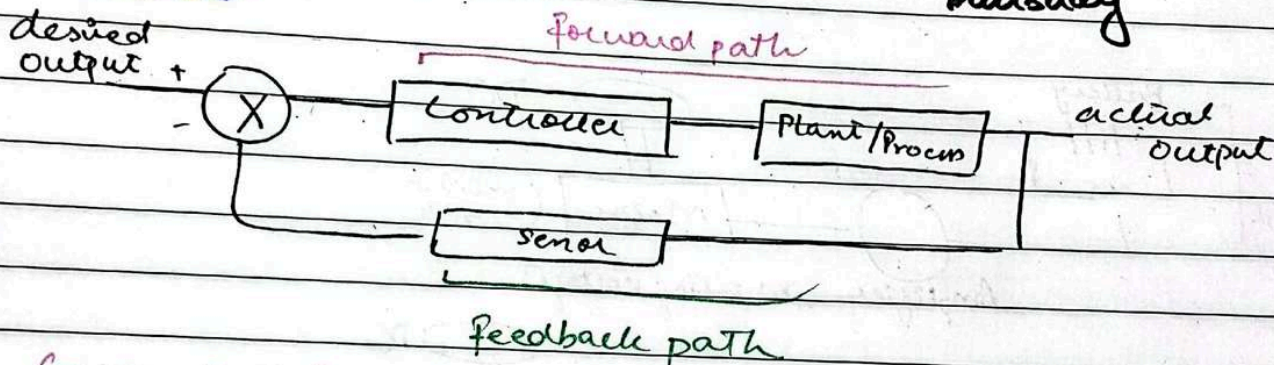
Forward path directing input to output

Feedback path the path in which output is connected back to comparator (summing point)
↳ Normally control sensor.

Lecture 2

23 Jan 2025

Thursday



Component of feedback

- plant/process
- controller
- comparator
- ↳ sensor

Time domain analysis (State pay (solve through matrix))

frequency domain analysis (Classical control Theory)

Every block should be represented by its transfer function

Transfer Function :

ratio of output over input in frequency domain

$x(s) \rightarrow \boxed{s} \rightarrow y(s)$

obtain by Laplace transform

$$T.F = \frac{Y(s)}{X(s)} = \frac{\text{Laplace transform of output}}{\text{Laplace transform of input}}$$

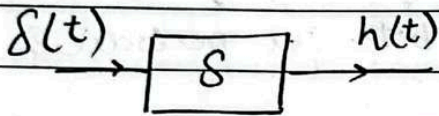
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Review of Laplace Transform

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$



We take Laplace inverse of impulse that give transfer function

$$h(t) \xrightarrow{\mathcal{L}} H(s) \text{ (transfer function)}$$

$$h(t) \xrightarrow{F} H(j\omega) \text{ (frequency response)}$$

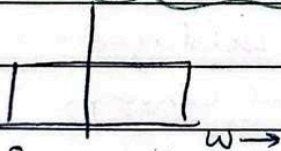
→ Transfer function of forward path is represented by $G(s)$

→ Feedback path transfer function is $H(s)$

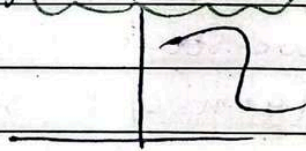
↳ just represent impulse function

How could we find the frequency response of signal?

We need to take Fourier transform for frequency response



ideal Lowpass filter (frequency response)



non ideal Low pass filter

→ We need Laplace Transform to find the transfer function of system

Can we find transfer function from impulse response of system?

Yes we can find if we take Laplace of impulse we will get transfer function

What is impulse response?

Output of the system when

input is unit impulse

Response is output of system

→ Every system we define the impulse response any type of system (R-network, amplifier) we can find impulse response that is a unique characteristic that tells the behaviour of system

→ If input is $\delta(t)$ output $h(t)$

→ Convolution give output of system (LTI system)

→ It give us zero state response (by keeping the initial condition zero)

→ Give the zero state response of LTI system provided the impulse response is given.



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→ To represent the signal as sum of sinusoid

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Fourier Transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

→ use for signal analysis

→ provide phase and magnitude spectrum (with respect of ω)

we came to know what frequency component are present in any random signal

Fourier Series → for periodic signal

Fourier Transform → can be perform both for periodic & aperiodic

Signal Analysis → Fourier Transform

System Analysis → Laplace Transform

Why we do not use Fourier analysis for representation of system?

Why we prefer Laplace over Fourier transform?

Condition for existence of Fourier transform

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

→ signal is absolutely integrable
→ signal is convergent

Dirichlet's Condition

$x = e^t$ no Fourier transform

$x = u(t)$ Fourier transform by $\text{sgn}(t)$

→ Fourier transform only exist for the signal that convergent
it is the disadvantage of Fourier transform

In system we need to cover the wider range of signal

→ bcz Fourier transform have strict condition on existence we cannot find Fourier transform of the signal that convergent

Reason 1 Laplace cover wider range of signal

Reason 2 You can convert differential equation in algebraic equation using Laplace transform

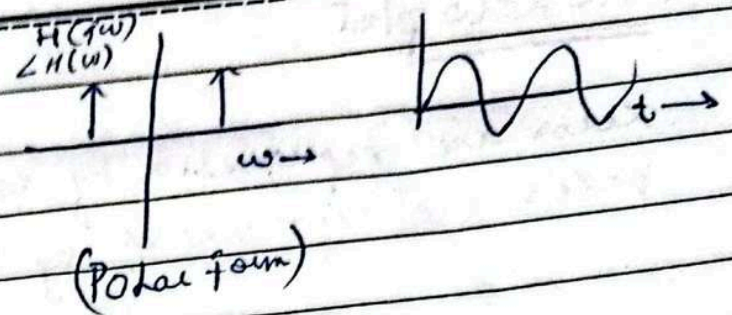


System Analysis → equation of system in some form

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$s = \sigma + j\omega$ (frequency domain)
 ↓ complex number in rectangular form
 ↓ sigma



s plane is a complex plane x axis (σ) y axis ($j\omega$)

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt$$

$$= \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt$$

if $\sigma = 0$ then it will be Fourier

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

→ Fourier transform is subset of Laplace
 → making real part zero

when we take Laplace transform we write signal in growing and decaying form

Poles and Zeros

$$H(s) = \frac{(s+2)(s+3)}{(s+1+2j)(s+1-2j)}$$

Zeros $s+2=0$ $s+3=0$

$$s = -2 \quad s = -3$$

Poles $s+1+2j=0$ $s+1-2j=0$

$$s = -1-2j \quad s = -1+2j$$

2 Zeros & 2 poles

Zeros are the point at which the function is equal to zero

→ numerator ko solve karay jo values ahey gi wo zero

Poles The value at which function is infinity

→ denominator ko zero ka equal karay ga or factor solve for order of s
 answer are location of plot



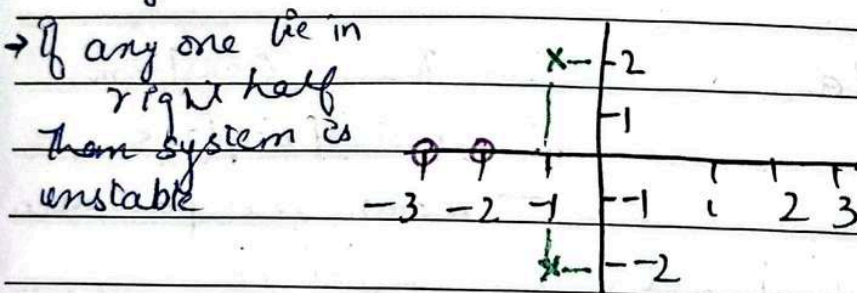
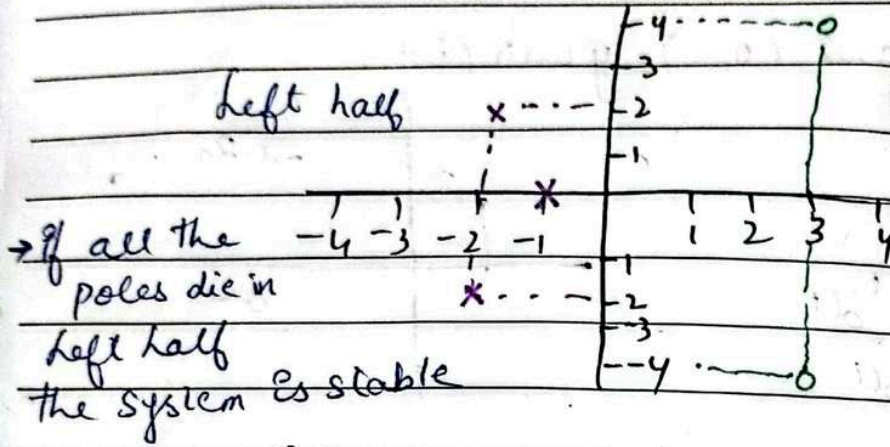
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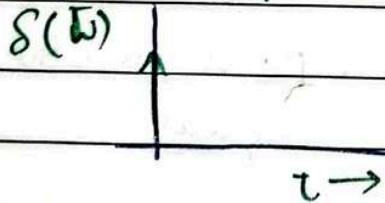
Pole zero plot

Zeros are represented by small circle
poles are represented by cross

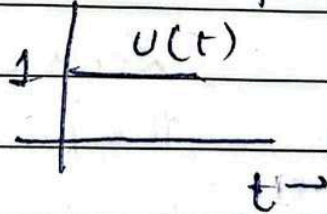
Poles are the point where function is Infinity
Zeros are point where transfer function is zero



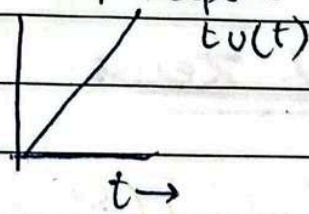
Impulse Response



Unit Response



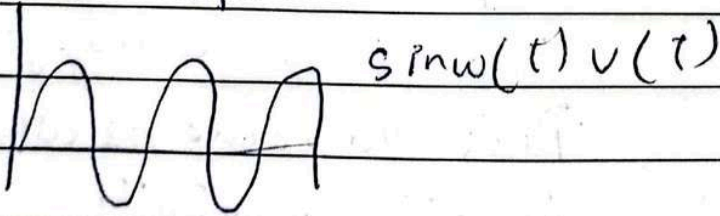
Ramp Response



$e^{-at}u(t)$

Exponential

Sinusoidal Response



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$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$x_1(t) = \delta(t)$$

$$x_1(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt$$

$$x_1(s) = e^{-s(0)} = 1$$

$$= \int \delta(t-2) e^{-st} dt = e^{-2s}$$

$$\delta(t-2)$$

↑

2

$$x_2(t) = u(t)$$

$$x_2(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \frac{1}{-s} \left[e^{-st} \right]_0^{\infty}$$

$$= \frac{1}{-s} \left[e^{-\infty} - e^0 \right]$$

$$x_2(s) = \frac{1}{s}$$

$$\delta(t) \xrightarrow{L} 1$$

$$u(t) \rightarrow \frac{1}{s}$$

$$t u(t) \rightarrow \frac{1}{s^2}$$

$$t^n u(t) \rightarrow \frac{n!}{s^{n+1}}$$

$$e^{-at} u(t) \rightarrow \frac{1}{s+a}$$

$$\sin \omega(t) u(t) \rightarrow \frac{\omega}{s^2 + \omega^2}$$

$$\cos \omega(t) u(t) \rightarrow \frac{s}{s^2 + \omega^2}$$

$$e^{-at} \sin \omega t u(t) \rightarrow \frac{\omega}{(s+a)^2 + \omega^2}$$

$$x_3(t) = t u(t)$$

$$x_3(s) = \int_{-\infty}^{\infty} t u(t) e^{-st} dt$$

$$= \int_0^{\infty} t e^{-st} dt$$

$$= \left. \frac{t e^{-st}}{s} \right|_0^{\infty} - \int_0^{\infty} \frac{1 e^{-st}}{-s} dt$$

$$= \frac{1}{s} \times \left. \frac{e^{-st}}{-s} \right|_0^{\infty}$$

$$= \frac{1}{s^2}$$



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$$x_4(t) = e^{-at} u(t)$$

$$X_4(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

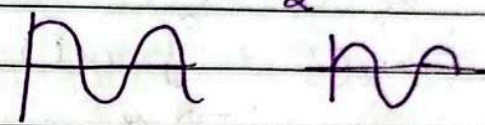
$$= \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_0^{\infty}$$

$$X_4(s) = \frac{1}{s+a}$$

Anti causal

$$\sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

$$\cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$



$$x_5(t) = \sin \omega t u(t)$$

$$= \frac{e^{j\omega t} - e^{-j\omega t}}{2j} u(t)$$

$$X_5(s) = \frac{1}{2j} \int_0^{\infty} e^{j\omega t} e^{-st} dt - \frac{1}{2j} \int_0^{\infty} e^{-j\omega t} e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} e^{-(s-j\omega)t} dt - \frac{1}{2j} \int_0^{\infty} e^{-(s+j\omega)t} dt$$

$$= \frac{1}{2j} \left[\left. \frac{e^{-(s-j\omega)t}}{-(s-j\omega)} \right|_0^{\infty} - \left. \frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \right|_0^{\infty} \right]$$

$$= \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right]$$

$$X_5(s) = \frac{\omega}{s^2 + \omega^2}$$



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Properties of Laplace Transform:

Linearity $a x_1(t) + b x_2(t) \longleftrightarrow a x_1(s) + b x_2(s)$

Time Shifting $x(t-t_0) \longleftrightarrow x(s)e^{-st_0}$

Frequency Shifting $e^{-at} x(t) \rightarrow X(s+a)$

Differentiation Property $\frac{d x(t)}{dt} \rightarrow s X(s) - x(0)$

$$\frac{d^2 x(t)}{dt^2} \rightarrow s^2 X(s) - s x(0) - x'(0)$$

$$\frac{d^n x(t)}{dt^n} \rightarrow s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k-1)}(0)$$

Integration Property

$$\int_0^t x(t) dt \rightarrow \frac{X(s)}{s}$$

